A Computer-Assisted Seven-Point Refinement for Simple Zeros of the Riemann Zeta Function

Anonymous submission
AI-assisted research draft

Peer-review draft, August 14, 2026

Status. This manuscript presents an unreviewed, computer-assisted proof of a refinement of Claude's Theorem D. It does not prove the Riemann hypothesis. The finite seven-point inequality has been replayed locally at two Arb precision settings, and its delicate strong-convexity basin has also been checked by a separate exact-rational LDL implementation using the same Arb/FLINT transcendental trust base. The endpoint deletion, finite-grid tail, uniform Gram-to-kernel comparison, and summation of all block errors are proved explicitly below. The prime-side trace asymptotics are cited from Claude's Theorem D [1]. A partial Lean 4 companion checks the finite-dimensional algebra but is not used as a substitute for the analytic proof. The result has not yet received independent specialist peer review.

Abstract

Let $N(T, 2T)$ count the nontrivial zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in the same interval. We retain a convex spectral defect in the rank–trace argument underlying Claude's 2026 Montgomery–Taylor bound and connect that defect to short-range correlations of consecutive critical-line zeros. The only new computer-assisted input is the global inequality

$$\mathcal{F}_6(g_1, \ldots, g_6) \geq C_* := \frac{38262312113}{10^{13}} \qquad (g_1, \ldots, g_6 \geq 0),$$

for an explicit seven-point functional built from the normalized Montgomery–Taylor overlap kernel. An exhaustive interval verifier uses Arb enclosures, certified convex tangent bounds, and two rational strong-convexity basins; a higher-precision replay has identical logical counts. A Lean companion formalizes the new matrix and exact-arithmetic layers and exact final arithmetic. We prove directly that endpoint deletion costs $o(N)$, that the finite simple-zero Gram entries differ uniformly from the limiting kernel by $O(1/\log T)$ on every relevant block, and that the error summed over all 267 shifts is $o(N)$. The Lean companion is supporting evidence rather than a premise of the proof. The resulting bound is

$$\liminf_{T\to\infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.67302547683787431811597399993\ldots.$$

1 Scope and statement of the result

Write

$$N := N(T, 2T), \qquad S := N_0^s(T, 2T),$$

and let

$$H_{\mathrm{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457343797908\ldots. \tag{1}$$

The quantity H_{MT} is the optimized constant in Theorem D of Claude [1]. That theorem is accompanied by a Lean 4 artifact for its stated results [2]. We use its analytic trace and tail estimates in the ordinary manner of a cited theorem. The accompanying `ZetaSeven` package is a pinned partial formalization of the new finite-dimensional layers; the mathematical proof in this manuscript does not depend on an end-to-end Lean theorem.

The exact local constant certified in this work is

$$C_* := \frac{38262312113}{10000000000000} = 0.0038262312113. \tag{2}$$

Theorem 1.1 (Computer-assisted refinement)**.** *One has*

$$\begin{aligned}\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} &\geq \frac{267H_{\mathrm{MT}} - 266/500}{267 - 261C_*} \\ &= \frac{2670000000000000H_{\mathrm{MT}} - 5320000000000}{2660013536538507} \\ &= 0.6730254768378743181159739993214 5\ldots.\end{aligned} \tag{3}$$

Equivalently, the lower proportion is $67.3025476837874318115\ldots\%$. *Relative to the cited Montgomery–Taylor value* $H_{\mathrm{MT}} = 67.2500703679411645\ldots\%$, *the increase is* $0.0524773158462672\ldots$ *percentage points.*

The theorem is unconditional: no Riemann hypothesis or zero-density hypothesis is assumed. As with any result built on a recent preprint, its status remains subject to checking of the cited source and independent review of the new argument. The adjective "computer-assisted" refers only to Proposition 5.1 and its explicit software trust base.

The proof has three new layers. Section 3 preserves a nonnegative spectral defect in the matrix inequality. Sections 4–5 lower-bound that defect using the Montgomery–Taylor kernel and a rigorous seven-point computation. Sections 6–7 convert the local inequality to the exact constant in Theorem 1.1.

2 The imported analytic interface

We state precisely the portion of [1] used by the refinement. Put

$$\ell := \log \frac{T}{2\pi}, \qquad L := \ell, \qquad I := (T, 2T], \qquad I' := (T - T^{1/2}, 2T + T^{1/2}].$$

Fix the ramp profile ϱ of [1, Section 2.2] and use ramp width one. The optimized window of [1, Section 7.1] is

$$\phi_L(u) := \cos\left(\frac{\sqrt{2}u}{L}\right)^{1/2} \varrho(L/2 - |u|), \qquad a_L := \frac{1}{L}\int_{\mathbb{R}} \phi_L(u)^2 \, du. \tag{4}$$

Put $h = 2\pi/L$, $d = \lfloor T/h \rfloor$, and $\tau_k = T + kh$. Let $\widehat{A} = A/(a_L L^2)$ be the normalized finite zero-side compression of Weil's Hermitian form from [1, Section 4]. Partition the zeros in I' into simple critical-line zeros $\mathcal{S}_1$, distinct multiple critical-line zeros $\mathcal{S}_2$, and unordered off-line pairs $\mathcal{P}$. Write

$$s_1 := \#\mathcal{S}_1, \qquad s_2 := \#\mathcal{S}_2, \qquad p := \#\mathcal{P}.$$

Counting multiplicity gives

$$N(I') \geq s_1 + 2s_2 + 2p. \tag{5}$$

For $\rho \in \mathcal{S}_1$, with ordinate γ_ρ, put

$$u_\rho := \big(\widehat{\phi}_L(\gamma_\rho - \tau_k)\big)_{0 \leq k < d}, \qquad v_\rho := \frac{u_\rho}{\sqrt{a_L}\, L},$$

and let V have these atoms as columns. The full-grid identity [1, Lemma 2.2] gives

$$\sum_{k \in \mathbb{Z}} |\widehat{\phi}_L(\gamma_\rho - \tau_k)|^2 = a_L L^2,$$

so every finite column v_ρ has norm at most one. Moreover

$$P_1 = VV^*, \qquad M = V^*V,$$

and for $Q' := \widehat{A} - P_1$ the inertia decomposition in the proof of [1, Proposition 4.4] gives

$$n_+(Q') \leq s_2 + p. \tag{6}$$

The trace and tail analysis in [1, Sections 4–7] yields the baseline constant (1). Corollary 3.2 below shows that the same estimates, after applying the strengthened matrix inequality, give

$$S \geq H_{\mathrm{MT}} N + \mathcal{D}(M^\circ) - o(N), \tag{7}$$

where M° is the Gram matrix of the retained central simple zeros and $\mathcal{D}$ is defined in (8) below.

The full-grid identity just quoted is exact. Its finite-grid truncation and the limiting kernel comparison are proved with explicit uniform error in Lemma 4.1; they are not assumed as an additional interface.

3 A stability refinement of the rank–trace inequality

Define the convex, nonnegative function

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t - 3, & t \geq 2, \end{cases} \qquad \mathcal{D}(M) := \operatorname{tr} \Psi(M) \tag{8}$$

for positive semidefinite M.

Lemma 3.1 (Stability-enhanced rank–trace inequality)**.** *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$, and let Q be Hermitian with $n_+(Q) \leq b$. Then*

$$\|P + Q\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 3r - 4b + \mathcal{D}(M). \tag{9}$$

Proof. Write $Q = Q_+ - Q_-$ with $Q_\pm \succeq 0$, $Q_+ Q_- = 0$, and $\operatorname{rank} Q_+ \leq b$. Since $P, Q_+ \succeq 0$,

$$\|P + Q\|_{\mathrm{F}}^2 \geq \|P - Q_-\|_{\mathrm{F}}^2 + \|Q_+\|_{\mathrm{F}}^2.$$

For every $q \geq 0$, $q^2 \geq 4q - 4$, hence

$$\|Q_+\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr} Q_+ - 4b. \tag{10}$$

Let $p_1 \geq \cdots \geq p_r \geq 0$ be the eigenvalues of M, padding the nonzero spectrum of P by zeros, and let $n_1 \geq n_2 \geq \cdots \geq 0$ be the eigenvalues of Q_-. Von Neumann's trace inequality gives $\operatorname{tr}(PQ_-) \leq \sum_{i \leq r} p_i n_i$. Therefore

$$\|P - Q_-\|_{\mathrm{F}}^2 + 4 \operatorname{tr} Q_- \geq \sum_{i \leq r} ((p_i - n_i)^2 + 4n_i).$$

For $p \geq 0$,

$$\min_{n \geq 0} ((p - n)^2 + 4n) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p). \tag{11}$$

It follows that

$$\|P - Q_-\|_{\mathrm{F}}^2 \geq 2 \operatorname{tr} P - r + \mathcal{D}(M) - 4 \operatorname{tr} Q_-. \tag{12}$$

Adding (10) and (12) gives

$$\|P + Q\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 2 \operatorname{tr} P - r - 4b + \mathcal{D}(M).$$

Finally $\operatorname{tr} P$ is the sum of the squared column norms of V, so $\operatorname{tr} P \leq r$. □

Corollary 3.2 (Global defect inequality)**.** *With the notation of Section 2,*

$$s_1 \geq 4 \operatorname{tr} \widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 - 2N(I') + \mathcal{D}(M). \tag{13}$$

After the endpoint truncation and analytic estimates of [1],

$$S \geq H_{\mathrm{MT}} N + \mathcal{D}(M^\circ) - o(N). \tag{14}$$

Proof. Apply Lemma 3.1 to $P_1 + Q' = \widehat{A}$. Equations (5) and (6) imply

$$\begin{aligned} \|\widehat{A}\|_{\mathrm{F}}^2 &\geq 4 \operatorname{tr} \widehat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \\ &\geq 4 \operatorname{tr} \widehat{A} - s_1 - 2N(I') + \mathcal{D}(M), \end{aligned}$$

which is (13).

Let M° be the principal block belonging to the simple zeros in $(T, 2T]$ that are at normalized distance at least L^2 from both endpoints. The number of zeros in $I' \setminus I$ is $O(T^{1/2} L) = o(N)$, and Lemma 4.1 below shows that the additional endpoint deletion costs $O(L^2) = o(N)$. Pinching M into M° and its complement gives $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$: pinching is an average of unitary conjugations, $X \mapsto \operatorname{tr} \Psi(X)$ is convex and unitarily invariant, and $\Psi \geq 0$.

In the notation of [1, Equation (4.4)], $\widehat{A} = \widehat{G} - \widehat{E}$. Proposition 4.2 there gives, at $\lambda = 1$,

$$\|\widehat{E}\|_1 \ll T^{-1/2}, \qquad \|\widehat{G}\|_{\mathrm{F}} \ll N^{1/2}.$$

Consequently

$$4 \operatorname{tr} \widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr} \widehat{G} - \|\widehat{G}\|_{\mathrm{F}}^2 - o(N).$$

The optimized-window form of Theorem 5.8 and the proof of Theorem D in [1] give

$$\operatorname{tr} \widehat{G} = N + o(N), \qquad \|\widehat{G}\|_{\mathrm{F}}^2 = \frac{1}{c_1^*} N + o(N),$$

where $2 - 1/c_1^* = H_{\mathrm{MT}}$. Also $N(I') = N + o(N)$. Substitution in (13), followed by deletion of the $o(N)$ noncentral points, yields (14). Remark 6.1 of [1] confirms that the endpoint $\lambda = 1$ used here is admissible; its relative source error is $O(\log \log T / \log T) = o(1)$. □

4 The Montgomery–Taylor overlap kernel

Let $h := 2\pi/L$ and normalize ordinates by

$$x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Lemma 4.1 (Uniform finite-Gram comparison)**.** *After deleting simple zeros in normalized endpoint strips of width L^2, the number deleted is $O(L^2) = o(N)$, and uniformly for all retained simple zeros,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + O(L^{-1}), \tag{15}$$

where

$$k(x) := \frac{K(x)}{K(0)}, \tag{16}$$

$$\begin{aligned} K(x) &:= \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\cos(2\pi xt)\,dt \\ &= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \qquad (17) \\ K(0) &= \sqrt{2}\sin(1/\sqrt{2}). \end{aligned}$$

The apparent singularities in (17) *are removable.*

Proof. Put

$$q_L(t) := \phi_L(Lt)^2, \qquad q(t) := \cos(\sqrt{2}t)\mathbf{1}_{[-1/2,1/2]}(t).$$

The two functions differ only on two intervals of total length $2/L$ and are bounded by one. Hence

$$\|q_L - q\|_1 \le \frac{2}{L}, \qquad |a_L - K(0)| \le \frac{2}{L}. \tag{18}$$

For large L, $a_L \ge K(0)/2$. Substitution $u = Lt$ in the full-grid identity of [1, Lemma 2.2], together with (18), gives uniformly for every real x,

$$\left| \frac{\displaystyle\int_{-1/2}^{1/2} q_L(t)e^{-2\pi ixt}\,dt}{a_L} - \frac{K(x)}{K(0)} \right| \ll L^{-1}. \tag{19}$$

It remains to truncate the full grid. The derivative bounds stated in the proof of [1, Theorem D] imply, with a constant depending only on the fixed ramp,

$$|\widehat{\phi}_L(r)| \le C_\varrho |r|^{-2} \qquad (r \ne 0), \tag{20}$$

uniformly in L. For a retained ordinate, the distance to every omitted grid point is at least

$$D_L := h(L^2 - 2) \ge \pi L$$

for all large L. On either side of the grid,

$$\sum_{j\ge 0}(D_L + jh)^{-4} \le D_L^{-4} + \frac{1}{3hD_L^3} = O(L^{-2}). \tag{21}$$

By Cauchy–Schwarz, (20), and (21), the omitted product tail is $O(L^{-2})$. After the Gram normalization by $a_L L^2$ it is $O(L^{-4})$. Combining this with (19) proves (15), uniformly in the two retained ordinates.

Finally, each deleted normalized strip has physical width $hL^2 = 2\pi L$. The local zero count $N(t+1) - N(t) \ll \log(t+3)$ gives $O(L \log T) = O(L^2) = o(N)$ deleted zeros. The elementary integral gives (17) and its value at zero. □

Set

$$w(x) := k(x)^2. \tag{22}$$

The verifier uses the equivalent entire expression

$$K(x) = \frac{1}{2}\left(\operatorname{sinc}(\pi x - 1/\sqrt{2}) + \operatorname{sinc}(\pi x + 1/\sqrt{2})\right), \tag{23}$$

where $\operatorname{sinc}(z) = \sin z / z$ with its continuous value at zero.

5 The certified seven-point inequality

For six nonnegative consecutive gaps define

$$\mathcal{F}_6(g_1, \dots, g_6) := \frac{1}{3000} \sum_{i=1}^{6} g_i + \sum_{r=1}^{6} \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \cdots + g_{i+r-1}). \tag{24}$$

The second term contains all 21 pair separations among seven ordered points.

Proposition 5.1 (Certified local inequality)**.** *For all $g_1, \dots, g_6 \geq 0$,*

$$\mathcal{F}_6(g_1, \dots, g_6) \geq C_* = \frac{38262312113}{10^{13}}. \tag{25}$$

Computer-assisted verification. The accompanying verifier uses `python-flint==0.8.0` and Arb interval arithmetic [3]. Its grid mesh is 1/4000. On every closed cell it evaluates (23), forms a rigorous lower bound for w, and widens conversion to IEEE-754 binary64 downward. A separate interval table lower-bounds w''. All later range queries use minima of those cell bounds.

The exact pressure cutoff is

$$q = \lceil C_* \cdot 4000 \cdot 3000 \rceil = 45915.$$

Thus $\sum_i g_i/3000$ proves the target whenever the sum of lower cell indices is at least q. Before forming six-dimensional boxes, the verifier also uses the one-variable contribution

$$U(g) := \frac{g}{3000} + \frac{1}{3} w(g).$$

The surviving cells form the three exact components

$$[3807, 4780], \qquad [7218, 9373], \qquad [10560, 44945].$$

Pressure filtering leaves 435 initial products of components.

For each box, every consecutive partial sum in (24) receives a range-minimum lower bound. If this direct interval bound is insufficient, the code lower-bounds the Hessian throughout the box using the w'' table. A binary64 LDL factorization is only a cheap filter; positive definiteness is rechecked with Arb. On a certified convex box, evaluation at its rational midpoint gives the rigorous tangent bound

$$\mathcal{F}_6(x) \geq \mathcal{F}_6(c) - \sum_{i=1}^{6} |\partial_i \mathcal{F}_6(c)| r_i.$$

Two neighborhoods of the numerical minimizers require a sharper local argument. The first has exact rational center represented by

$$\begin{aligned} c = (&1.04608035577,\ 1.98913202062,\ 1.98641493611, \\ &1.04160329372,\ 1.97702352234,\ 1.04500209462). \end{aligned} \tag{26}$$

and the second has reversed coordinates. Each coordinate radius is exactly 1/64. Interval lower bounds for w'', followed by an Arb LDL check, prove throughout both basins that

$$\nabla^2 \mathcal{F}_6 \succeq \frac{3}{16} I. \tag{27}$$

At 256-bit basin precision, strong convexity gives

$$\begin{aligned} \inf_{\text{basin}} \mathcal{F}_6 &\geq \mathcal{F}_6(c) - \frac{\|\nabla \mathcal{F}_6(c)\|^2}{2(3/16)} \\ &\geq 0.0038262312113044742482737171300627069938382303\,5, \end{aligned} \tag{28}$$

and the same lower endpoint for the reversed basin. This clears the exact target by more than 4.47×10^{-15}. A search box is basin-pruned only when all of its closed grid cells lie inside a certified basin.

All other boxes are bisected along a widest coordinate. Reaching an unresolved terminal cell raises an exception. The canonical run has the following deterministic counts.

Field	Canonical value
Initial boxes	435
Visited nodes	822,433
Pruned leaves	411,434
Splits	410,999
Maximum depth	39
Interval prunes	285,258
Pressure prunes	2,872
Tangent prunes	122,329
Strong-convexity prunes	975
Unresolved terminal cells	0

The code checks both binary-forest identities

$$822433 = 411434 + 410999, \qquad 411434 = 410999 + 435.$$

The canonical kernel-table SHA-256 is
`5a1f95f754a83ba05f37692d4bda69bf3fa5d3752af902826bfc4cffd31428b9`,
and the second-derivative-table SHA-256 is
`318ee79f619cc75c6f7a30424a181764865910693f33a3d95469ebcae0ce42ba`.

A full rerun with table precision increased from 128 to 160 bits and basin precision from 256 to 320 bits produces the same initial boxes, node counts, pruning counts, and depth. Its independently regenerated second-derivative table has SHA-256 fd634c87441380105aad52f069f735108d0dabcd7d5d0a993b6abc52ee12b055.

This is a cross-precision stress test, not an independent implementation: it uses the same source and Arb/FLINT library. The exact expected reports are included in the artifact and are compared field by field on every release run. At commit

b8dcb42c88f920fca2c609b1c88a5cc133d88b2a,

clean local replays with the hash-locked CPython 3.12 wheel ended respectively with

`release_certificate=match,` `cross_precision_certificate=match.`

The additional script `verifier/scripts/verify_basin_independent.py` does not import the release verifier. It reimplements the sinc quotient derivatives, subdivides each scalar basin interval into 2,048 exact rational pieces, and performs the shifted 6×6 LDL test over exact rational arithmetic. It proves (27) for both basins and obtains the lower endpoint

$$0.0038262312113044742482737171300627069938382303548849123 86.$$

This is a separate control path but retains Arb/FLINT as the transcendental interval trust base. These facts complete the finite verification of Proposition 5.1. ☐

Remark 5.2. *The center* (26) *was located by ordinary floating-point optimization, but the optimizer output is not trusted. Only the displayed terminating decimals, interpreted as exact rationals, and the interval proof of* (27)–(28) *enter the certificate.*

6 From seven-point windows to block defect

Let $y_1 < \cdots < y_m$ and set

$$E_m := 2 \sum_{1\le i<j\le m} w(y_j - y_i). \tag{29}$$

Lemma 6.1 (Block energy)**.** *For every $m \ge 7$,*

$$E_m + \frac{1}{500}(y_m - y_1) \ge C_*(m-6). \tag{30}$$

Proof. Write $g_i = y_{i+1} - y_i$ and sum (25) over the $m-6$ consecutive seven-point windows. A pair spanning $r \le 6$ gaps occurs in at most $7-r$ windows and has coefficient $2/(7-r)$ in each, so its total coefficient is at most 2. Pairs spanning more than six gaps do not appear and contribute nonnegatively to E_m. Each single gap occurs in at most six windows, so its total linear coefficient is at most $6/3000 = 1/500$. ☐

Lemma 6.2 (Block defect)**.** *If $G \succeq 0$ is Hermitian, then*

$$\operatorname{tr}\Psi(G) \ge \min\left\{1, 2\sum_{i<j} |G_{ij}|^2\right\}. \tag{31}$$

Proof. If every eigenvalue of G is at most 2, then $\Psi(G) = (G-I)^2$ and

$$\operatorname{tr}\Psi(G) = \operatorname{tr}(G-I)^2 \ge 2\sum_{i<j}|G_{ij}|^2.$$

If some eigenvalue $\lambda > 2$, then $\Psi(\lambda) = 2\lambda - 3 > 1$, while all other contributions are nonnegative. ☐

Choose

$$m := 267, \qquad A_0 := C_*(m-6) = 0.9986463461493 < 1. \tag{32}$$

For comparison,

$$C_*(268-6) = 1.0024725773606 > 1. \tag{33}$$

Thus $m = 267$ is the last block size before the unit cap in Lemma 6.2 activates. More explicitly, for general m the block constant is $A_m = \min\{1, C_*(m-6)\}$ and the same argument gives

$$R_m = \frac{H_{\mathrm{MT}} - (m-1)/(500m)}{1 - A_m/m}.$$

On $7 \le m \le 267$ this is increasing because $6C_*(H_{\mathrm{MT}} - 1/500) > (1 - C_*)/500$, while for $m \ge 268$ it is decreasing with derivative $-H_{\mathrm{MT}}/(m-1)^2$ after writing $R_m = ((H_{\mathrm{MT}} - 1/500)m + 1/500)/(m-1)$. Hence $m = 267$ maximizes the resulting bound among integer block sizes $m \ge 7$.

Let B be a block of 267 consecutive retained central simple zeros, and let G_B be its Gram matrix. Lemma 4.1 and the bounds $|(G_B)_{ij}|, |k(x_i - x_j)| \le 1$ give, uniformly in B,

$$\left| 2\sum_{i<j} |(G_B)_{ij}|^2 - E_m \right| \le \frac{C_m}{L}, \tag{34}$$

where C_m is absolute because $m = 267$ is fixed. Put $Q_B = 2\sum_{i<j} |(G_B)_{ij}|^2$. If $Q_B \le 1$, then Lemmas 6.1–6.2 and (34) give the next inequality. If $Q_B > 1$, then $\mathcal{D}(G_B) \ge 1 > A_0$, so it holds again. Thus, uniformly in B,

$$\mathcal{D}(G_B) + \frac{1}{500}\,\mathrm{span}(B) \ge A_0 - O(L^{-1}). \tag{35}$$

7 Shifted block pinching and final arithmetic

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalized ordinates of the retained central simple zeros. Endpoint deletion gives

$$S^\circ = S - O(L^2) = S - o(N). \tag{36}$$

For each of the $m = 267$ offsets, partition the ordered points into full consecutive blocks of size m, leaving only bounded endpoint remainders. Pinching M° to the principal blocks and the remainder gives

$$\mathcal{D}(M^\circ) \ge \sum_B \mathcal{D}(G_B). \tag{37}$$

For each offset, sum (35) over its full blocks and use (37); then average over all m offsets. The average number of full blocks is $S^\circ/m + O(1)$. Each adjacent gap belongs to a block span for at most $m-1$ offsets. Moreover

$$x_{S^\circ} - x_1 \le \frac{LT}{2\pi} = N + o(N) \tag{38}$$

by the Riemann–von Mangoldt formula. Across all offsets there are at most $S^\circ \le N$ full blocks, each occurring once, so the total error from (35) is $O(N/L) = o(N)$. Consequently

$$\mathcal{D}(M^\circ) \ge \frac{A_0}{m}S - \frac{m-1}{500m}N - o(N) = \frac{261C_*}{267}S - \frac{266}{133500}N - o(N). \tag{39}$$

Substitute (39) into (14):

$$S \geq H_{\mathrm{MT}} N + \frac{261 C_*}{267} S - \frac{266}{133500} N - o(N).$$

Hence

$$\left(1 - \frac{261 C_*}{267}\right) S \geq \left(H_{\mathrm{MT}} - \frac{266}{133500}\right) N - o(N). \tag{40}$$

Divide by N, let $T \to \infty$, and use (2). This gives (3). Multiplying numerator and denominator by $267 \cdot 10^{13}$ gives the exact reduced expression in Theorem 1.1; the common integer gcd is one. This proves Theorem 1.1.

8 Position relative to prior work

Montgomery's pair-correlation argument proved, under RH, that at least two thirds of the zeros are simple [4]; the optimized Montgomery–Taylor window gives the constant (1) [5]. Chirre, Gonçalves, and de Laat later obtained stronger conditional simple-zero estimates by semidefinite programming and positivity outside the bandwidth-one interval [6]. Those results operate in a different conditional regime and are not numerically comparable to Theorem 1.1 as an unconditional critical-line statement.

Recent work of Goldston and collaborators clarified how pair correlation can control zeros that are simultaneously simple and on the critical line, under the pair-correlation conjecture or narrow-box hypotheses [7, 8, 9]. Claude's paper [1] then supplied the unconditional finite-dimensional inertia framework and the baseline H_{MT} used here. The present refinement does not extend Fourier support or add a new prime-side estimate. It extracts extra information from the short-range Gram matrix by preserving the spectral defect $\mathcal{D}$ and certifying a seven-point local inequality.

The software lineage is also relevant. A public AI-generated artifact first implemented the stability approach at local target 0.0038 [10]; a clean-room release certified 0.00382 [11]. The current artifact certifies (2) and adds the explicit strong-convexity basins and precision replay. These repositories are research artifacts, not substitutes for peer-reviewed priority analysis. A search of the cited 2025–2026 primary literature found no identical stability-defect/seven-point construction, but that search cannot establish priority.

9 Reproducibility, trust base, and limitations

Proposition 5.1 is the only computer-assisted proposition in the deduction. The accompanying source archive contains the verifier, exact JSON reports, unit tests, a hash-locked dependency file, and detailed audit and reproduction notes. The canonical commands are

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --require-hashes -r requirements.lock
PYTHONPATH=src python -m unittest discover -s tests -v
PYTHONPATH=src python scripts/verify_release.py
PYTHONPATH=src python scripts/verify_cross_precision.py
python scripts/verify_basin_independent.py
```

The two verifier scripts agree with their expected reports when they end with the respective status strings

```
release_certificate=match
cross_precision_certificate=match.
```

The partial Lean companion uses Lean `v4.33.0-rc2` and is pinned to the following upstream and mathlib commits:

```
upstream:  3635e74826a4c1fcece7d1cd2b6fa75e43a00510
mathlib:  51e6992efd06126df61a496bebf8f49482a4e129.
```

Its audit commands are

```
lake build ZetaSeven
lake env lean ZetaSeven/Audit.lean
```

The proved declarations cover the rank–trace surplus, the simple-zero Gram split and its inertia bound, the finite tail seam, the bridge from off-diagonal Gram energy to spectral defect, the exact consecutive-window count, arbitrary principal-block spectral pinching, abstract 267-offset aggregation, the conditional 267-point block bridge, the abstract defect-preserving Theorem D source inequality (including an explicit $o(N)$ error package), the concrete shifted partitions and endpoint fibers, their insertion into the exact finite simple-zero count, and the exact rational and epsilon-form assembly. The proposition `SevenPointClaim` is intentionally defined but not asserted as a theorem. Consequently, a successful Lean build is evidence for these layers only; it is not an end-to-end formal proof of Theorem 1.1.

The finite trust base consists of CPython 3.12, IEEE-754 binary64 semantics, `python-flint` version 0.8.0, Arb/FLINT, the operating system and hardware, and the included source. The verifier rebuilds all transcendental tables; it does not trust cached tables, the numerical optimizer, or committed run logs. The separate basin checker changes the derivative implementation, subdivision, and exact linear-algebra path, but uses the same Arb/FLINT library. No fully independent interval library or proof-assistant checker is yet available for the complete six-variable subdivision. Lean reduces the trust base for the surrounding algebra but does not reduce the Arb certificate's trust base until a proof-carrying certificate checker is completed.

The larger mathematical trust boundary is equally important. This work does not independently reprove the prime-side Theorem 5.8 of [1] or its explicit-formula inputs. Corollary 3.2 and Lemma 4.1 spell out the new interface, endpoint deletion, finite-grid tail, and uniform Gram comparison rather than leaving them as formal hypotheses. Reviewers should nevertheless audit their normalization against the upstream concrete parameter family and independently assess the still-new cited theorem.

For peer review, the highest-value checks are:

1. verify Lemma 3.1 and the deduction of Corollary 3.2 against the exact notation of [1];
2. independently inspect the cell-covering, outward-rounding, Hessian, and basin-containment logic in the verifier;
3. reproduce both certificates on a clean machine and, preferably, a second architecture;
4. implement an independent interval or proof-assistant checker for Proposition 5.1;
5. check the seven-window multiplicities and shifted gap charges in Sections 6–7;
6. repeat the prior-art search before any public priority claim.

Finally, neither a positive proportion result nor the local statistics used here imply that all zeros lie on the critical line. The method remains far from a proof of the Riemann hypothesis, and the bandwidth-one framework has known structural ceilings discussed in [1].

Acknowledgment and authorship placeholder

This anonymous line is intended for review circulation only. A submission must name the people who take responsibility for the mathematics, disclose the AI-assisted origin in accordance with the venue's policy, and obtain their approval of the final text. The upstream software credits and license are recorded in the accompanying `PROVENANCE.md`.

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